Jingyu (Jack) Zhang

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RESEARCH INTERESTS

My research focuses on post-training and alignment for LLM agents, with an emphasis on ensuring safe and reliable model behavior. I study training and evaluation methods that enable robust generalization through reasoning over behavioral specifications and learning from interaction. Along these lines, my recent work includes reinforcement learning for self-improving alignment, multi-agent collaboration, enhancing safety controllability, and stress-testing instruction hierarchy as complexity scales.

EDUCATION

Johns Hopkins University **Aug 2023 - Present**
Ph.D. in Computer Science *Expected Graduation: January 2027*
Advisors: Daniel Khashabi and Benjamin Van Durme

Johns Hopkins University **Aug 2019 - May 2023**
B.S. in Computer Science *GPA: 3.97/4.00*
Additional Majors: Mathematics; Applied Mathematics & Statistics. Minor: Economics

RESEARCH EXPERIENCE

Apple AIML - Apple Foundation Model Team *June - Sep 2026*
Research Intern
Mentors: Joseph Yitan Cheng and Shruti Palaskar

MiniMax - Agents Team *Jan - Mar 2026*
Member of Technical Staff Intern
Mentors: Chi Zhang and Pengyu Zhao

Meta Superintelligence Labs - Trust Alignment Team *May - Dec 2025 (P/T after Aug 2025)*
Research Scientist Intern
Mentors: Hongyuan Zhan and Jason Weston

Microsoft - Responsible AI Research *May 2024 - Apr 2025 (P/T after Aug 2024)*
Research Intern
Mentor: Ahmed Elgohary Ghoneim

University of Washington *Summer 2022 - Spring 2023*
Research Intern
Advisors: Yulia Tsvetkov and Tianxing He

MIT Computer Science and Artificial Intelligence Laboratory *Spring 2022 - Summer 2022*
Research Intern
Advisors: James Glass and Tianxing He

PUBLICATIONS & PREPRINTS

Selected & Recent Works

1. **SIGMA: Self-Improving Alignment Generalization from a Model Spec** [Paper]
   ***Jingyu Zhang**, Shruti Palaskar, Daniel Khashabi, Benjamin Van Durme, Leon Gatys, Joseph Yitan Cheng.*
   ICLR 2027 submission.
2. **The Alignment Waltz: Jointly Training Agents to Collaborate for Safety** [Paper]
   ***Jingyu Zhang**, Haozhu Wang, Eric Michael Smith, Sid Wang, Amr Sharaf, Mahesh Pasupuleti, Benjamin Van Durme, Daniel Khashabi, Jason Weston, Hongyuan Zhan.*
   ICLR 2026.
3. **Controllable Safety Alignment: Inference-Time Adaptation to Diverse Safety Requirements** [Paper]
   ***Jingyu Zhang**, Ahmed Elgohary, Ahmed Magooda, Daniel Khashabi, Benjamin Van Durme.*
   ICLR 2025.

4. **Many-Tier Instruction Hierarchy in LLM Agents** [Paper]
   Jingyu Zhang, *Tianjian Li, William Jurayj, Hongyuan Zhan, Benjamin Van Durme, Daniel Khashabi.*
   EMNLP 2026 Findings.
5. **Verifiable by Design: Aligning Language Models to Quote from Pre-Training Data** [Paper]
   Jingyu Zhang, *Marc Marone, Tianjian Li, Benjamin Van Durme, Daniel Khashabi.*
   NAACL 2025 (oral presentation).
6. **SemStamp: A Semantic Watermark with Paraphrastic Robustness for Text Generation** [Paper]
   Abe Bohan Hou, **Jingyu Zhang***, Tianxing He*, Yichen Wang, Yung-Sung Chuang, Hongwei Wang, Lingfeng Shen, Benjamin Van Durme, Daniel Khashabi, Yulia Tsvetkov.*
   NAACL 2024. [*Equal Contribution]

All Papers

7. **Self-Compacting Language Model Agents** [Paper]
   *Tianjian Li, **Jingyu Zhang**, William Jurayj, Xi Wang, Chuanyang Jin, Mehrdad Farajtabar, Eric Nalisnick, Daniel Khashabi.*
   NeurIPS 2026.
8. **Minimally Invasive Steering of Language Models** [Paper]
   *Taha Entesari, **Jingyu Zhang**, Daniel Khashabi, Mahyar Fazlyab.*
   NeurIPS 2026.
9. **The MiniMax-M2 Series: Mini Activations Unleashing Max Real-World Intelligence** [Paper]
   *Aili Chen, Aonian Li, Baichuan Zhou, Bangwei Gong, ..., **Jingyu Zhang**, ... (218 authors).*
   Technical Report.
10. **Prioritization of Risks from Artificial Intelligence: A Delphi Study of 272 International Experts** [Paper]
    *Alexander K. Saeri, Jess Graham, Michael Noetel, Peter Slattery, ..., **Jingyu Zhang**, ... (188 authors).*
    arXiv Preprint.
11. **Accurate but Not Humble: Evaluating Epistemic Humility in LLM Agents under Knowledge Conflict** [Paper]
    *Kaiser Sun, Bernal Jiménez Gutiérrez, Hongjun Liu, **Jingyu Zhang**, Jie Gao, Mark Dredze, Daniel Khashabi.*
    EMNLP 2026 Findings.
12. **DeonticBench: A Benchmark for Reasoning over Rules** [Paper]
    *Guangyao Dou, Luis Brena, Akhil Deo, William Jurayj, **Jingyu Zhang**, Nils Holzenberger, Benjamin Van Durme.*
    AAAI 2027 submission.
13. **Reasoning over Mathematical Objects: On-Policy Reward Modeling and Test Time Aggregation** [Paper]
    *Pranjal Aggarwal, Marjan Ghazvininejad, Seungone Kim, Ilia Kulikov, Jack Lanchantin, Xian Li, Tianjian Li, Bo Liu, Graham Neubig, Anaelia Ovalle, Swarnadeep Saha, Sainbayar Sukhbaatar, Sean Welleck, Jason Weston, Chenxi Whitehouse, Adina Williams, Jing Xu, Ping Yu, Weizhe Yuan, **Jingyu Zhang**, Wenting Zhao.*
    arXiv Preprint.
14. **Are Finer Citations Always Better? Rethinking Granularity for Attributed Generation** [Paper]
    *Hexuan Wang, **Jingyu Zhang**, Benjamin Van Durme, Daniel Khashabi.*
    ACL ARR submission.
15. **Beyond Reasoning Gains: Mitigating General-Capability Forgetting in Large Reasoning Models** [Paper]
    *Hoang Phan, Xianjun Yang, Kevin Yao, **Jingyu Zhang**, Shengjie Bi, Xiaocheng Tang, Madian Khabsa, Lijuan Liu, Deren Lei.*
    ACL 2026 Findings.
16. **Jailbreak Distillation: Renewable Safety Benchmarking** [Paper] [Press]
    Jingyu Zhang, *Ahmed Elgohary, Xiawei Wang, A S M Iftekhar, Ahmed Magooda, Benjamin Van Durme, Daniel Khashabi, Kyle Jackson.*
    EMNLP 2025 Findings.
17. **Certified Mitigation of Worst-Case LLM Copyright Infringement** [Paper]
    Jingyu Zhang, *Jiacan Yu, Marc Marone, Benjamin Van Durme, Daniel Khashabi.*
    EMNLP 2025.

18. **Can A Society of Generative Agents Simulate Human Behavior and Inform Public Health Policy? A Case Study on Vaccine Hesitancy** [Paper]
Abe Bohan Hou, Hongru Du, Yichen Wang, ***Jingyu Zhang****, Zixiao Wang, Paul Pu Liang, Daniel Khashabi, Lauren Gardner, Tianxing He.*
COLM 2025.

19. **Rationalyst: Mining Implicit Rationales for Process Supervision of Reasoning** [Paper]
Dongwei Jiang, Guoxuan Wang, Yining Lu, Andrew Wang, ***Jingyu Zhang,*** *Chuyu Liu, Benjamin Van Durme, Daniel Khashabi.*
ACL 2025.

20. **Core: Robust Factual Precision with Informative Sub-Claim Identification** [Paper]
Zhengping Jiang, ***Jingyu Zhang****, Nathaniel Weir, Seth Ebner, Miriam Wanner, Kate Sanders, Daniel Khashabi, Anqi Liu and Benjamin Van Durme.*
ACL 2025 Findings.

21. **Self-[In]Correct: LLMs Struggle with Discriminating Self-Generated Responses** [Paper]
Dongwei Jiang, ***Jingyu Zhang****, Orion Weller, Nathaniel Weir, Benjamin Van Durme, Daniel Khashabi.*
AAAI 2025.

22. **Tur[k]ingBench: A Challenge Benchmark for Web Agents** [Paper]
Kevin Xu, Yeganeh Kordi, Kate Sanders, Yizhong Wang, Adam Byerly, ***Jingyu Zhang****, Benjamin Van Durme, Daniel Khashabi.*
NAACL 2025.

23. **DiffNorm: Self-Supervised Normalization for Non-autoregressive Speech-to-speech Translation** [Paper]
Weiting Tan, ***Jingyu Zhang****, Lingfeng Shen, Daniel Khashabi, Philipp Koehn.*
NeurIPS 2024.

24. **k-SemStamp: A Clustering-Based Semantic Watermark for Detection of Machine-Generated Text** [Paper]
Abe Bohan Hou, ***Jingyu Zhang****, Yichen Wang, Daniel Khashabi, Tianxing He.*
ACL 2024 Findings.

25. **The Language Barrier: Dissecting Safety Challenges of LLMs in Multilingual Contexts** [Paper]
Lingfeng Shen, Weiting Tan, Sihao Chen, Yunmo Chen, ***Jingyu Zhang****, Haoran Xu, Boyuan Zheng, Philipp Koehn, Daniel Khashabi.*
ACL 2024 Findings.

26. **On the Zero-Shot Generalization of Machine-Generated Text Detectors** [Paper]
Xiao Pu, ***Jingyu Zhang****, Xiaochuang Han, Yulia Tsvetkov, Tianxing He.*
EMNLP 2023 Findings.

27. **On the Blind Spots of Model-Based Evaluation Metrics for Text Generation** [Paper]
Tianxing He,* ***Jingyu Zhang*****, Tianle Wang, Sachin Kumar, Kyunghyun Cho, James Glass, Yulia Tsvetkov.*
ACL 2023 (oral presentation). [*Equal Contribution]

28. **Geo-Seq2seq: Twitter User Geolocation on Noisy Data through Sequence to Sequence Learning** [Paper]
Jingyu Zhang*, Alexandra DeLucia, Chenyu Zhang, Mark Dredze.*
ACL 2023 Findings.

29. **PCFG-based Natural Language Interface Improves Generalization for Controlled Text Generation** [Paper]
Jingyu Zhang*, James Glass, Tianxing He.*
SEM 2023.* Preliminary version accepted at *2nd Workshop on Efficient Natural Language and Speech Processing (ENLSP), NeurIPS 2022.* ***Best Paper Award**.*

30. **Changes in Tweet Geolocation over Time: A Study with Carmen 2.0** [Paper]
Jingyu Zhang*, Alexandra DeLucia, Mark Dredze.*
8th Workshop on Noisy User-generated Text (W-NUT), COLING 2022.

31. **Study of Manifestation of Civil Unrest on Twitter** [Paper]
Abhinav Chinta,* ***Jingyu Zhang*****, Alexandra DeLucia, Anna L. Buczak, Mark Dredze.*
7th Workshop on Noisy User-generated Text (W-NUT), EMNLP 2021. [*Equal Contribution]

SELECTED AWARDS

- **Amazon AI Fellowship** - $100,000 stipend + tuition/fees over two years. Press coverage: .
- **ICLR 2026 Travel Grant** - $1,600 + conference registration.
- **Thinking Machine Labs Tinker Research Grant** - $5,000 in Tinker API credits.
- **Michael J. Muuss Research Award** - Top 1 out of 744 JHU CS undergraduates (2023). Press coverage: .
- **Best Paper Award** - ENLSP Workshop at NeurIPS 2022 ($1,000 prize).
- **Pistritto Research Fellowship** - $4,000 grant (Fall 2022). Press coverage: .
- **Bloomberg Distinguished Professor (BDP) Summer Program Recipient** - $6,000 grant (Summer 2021).
- **National Olympiad in Informatics in Provinces (NOIP)** - National 1st Prize Certification (2018).

INDUSTRY EXPERIENCE

Apple Inc. **Research Intern**
Apple Foundation Model Team *Jun 2026 - Sep 2026*

- Conducted research on self-improving alignment and enhancing alignment generalization (Paper submitted to ICLR 2027).
- Contributed to safety alignment recipe for Apple Foundation Model.

MiniMax **Member of Technical Staff Intern**
Agents Team *Jan 2026 - Mar 2026*

- Conducted research on hallucination detection and benchmarking in agentic environments.
- Contributor to the MiniMax-M2 series of models (technical report: `https://arxiv.org/abs/2605.26494`).

Meta Platforms, Inc. **Research Scientist Intern**
Meta Superintelligence Labs - Trust Alignment *May 2025 - Dec 2025 (P/T after Aug 2025)*

- Conducted research related to multi-agent safety alignment (Paper published in ICLR 2026: `https://arxiv.org/abs/2510.08240`).
- Implemented a multi-agent reinforcement learning framework to support model co-evolution.
- Collaborated with FAIR on enhancing the parallel reasoning capabilities for LLMs (paper: `https://arxiv.org/abs/2603.18886`; blog post: `https://facebookresearch.github.io/RAM/blogs/paragator/`).

Microsoft Corporation **Research Intern**
Responsible AI Research *May 2024 - Apr 2025 (P/T after Aug 2024)*

- Conducted research related to safety alignment and automatic red teaming.
- Resulted in two papers: Controllable Safety Alignment (ICLR 2025, `https://arxiv.org/abs/2410.08968`), Jailbreak Distillation (EMNLP 2025 Findings, `https://arxiv.org/abs/2505.22037`).

ByteDance Ltd. **C++ Development Intern**
Lark Explorer Department *May 2020 - May 2021 (P/T after Aug 2020)*

- Carried out C++ cross-platform development interacting with macOS kernel and Windows Win32 API.
- Performed client-side development with Electron and Node.js related to performance optimization.
- Conducted data science analytics on extensive user-generated data with Apache Hive and Python.

INVITED TALKS

The Alignment Waltz: Jointly Training Agents to Collaborate for Safety
Invited talk, CLSP Seminar, Johns Hopkins University, Baltimore, Maryland Oct 2025
Invited talk, OSU NLP Group, Ohio State University, Columbus, Ohio Nov 2025
Poster presentation, ICLR 2026, Rio de Janeiro, Brazil Apr 2026

Jailbreak Distillation: Renewable Safety Benchmarking
Invited talk, Microsoft Responsible AI Tech Talks, Redmond, Washington May 2025
Poster presentation, EMNLP 2025, Suzhou, China Nov 2025

Controllable Safety Alignment: Inference-Time Adaptation to Diverse Safety Requirements
Invited talk, CLSP Seminar, Johns Hopkins University, Baltimore, Maryland Oct 2024
Invited talk, Microsoft Responsible AI Tech Talks, Redmond, Washington Dec 2024
Invited talk, ZhiDX AI Talk, Online June 2025

Verifiable by Design: Aligning Language Models to Quote from Pre-Training Data
Oral presentation, the 11th Mid-Atlantic Student Colloquium on Speech, Language and Learning May 2024
Invited talk, Microsoft Hallucination Deep Dive, Redmond, Washington Aug 2024
Invited talk, Institute for Assured Autonomy, Johns Hopkins University, Baltimore, Maryland Oct 2024
Oral presentation, NAACL 2025, Albuquerque, New Mexico May 2025

SemStamp: A Semantic Watermark with Paraphrastic Robustness for Text Generation
Invited talk, CLSP Seminar, Johns Hopkins University, Baltimore, Maryland Dec 2023

On the Blind Spots of Model-Based Evaluation Metrics for Text Generation
Oral Presentation, ACL 2023, Toronto, Canada July 2023

PCFG-based Natural Language Interface Improves Generalization for Controlled Text Generation
Spotlight Presentation (Best Paper Award), ENLSP Workshop, NeurIPS 2022, New Orleans, USA Dec 2022

TEACHING & MENTORSHIP EXPERIENCE

EN.601.471/671 NLP: Self-supervised Models **Head Teaching Assistant**
JHU Department of Computer Science *Fall 2026*
- EN.601.471/671 is a mixed graduate / upper-level undergraduate course taught by Prof. Daniel Khashabi.
- Organized and conducted grading of quizzes and office hours on a weekly basis.

EN.601.465/665 Natural Language Processing **Course Assistant**
JHU Department of Computer Science *Fall 2021, Fall 2022*
- EN.601.465/665 is a mixed graduate / upper-level undergraduate course in NLP taught by Prof. Jason Eisner.
- Conducted grading of homework and exam papers and held review sessions and office hours on a weekly basis.
- Received an average score of 4.80/5.00 on student TA evaluation (100% "Good" or "Excellent" rating).

Code in Place 2021 **Section Leader (Volunteer)**
Stanford University Department of Computer Science *April 2021 - May 2021*
- Worked with a team of more than 50 teaching leads and 1000 section leaders to support 10,000+ students across the world as they navigate the first five weeks of CS 106A: Programming Methodology course.
- Prepared materials and taught a Python programming section of 10 students on a weekly basis.

Students Mentored
- Abe Bohan Hou, undergraduate student at JHU. Now PhD student at Stanford University.
- Matt Hexuan Wang, undergraduate student at JHU.
- Jiacan Yu, master's student at JHU.

ACADEMIC SERVICE

- Application Mentor, JHU CLSP pre-application support program (2023-present)
- Curriculum Committee, Department of Computer Science, Johns Hopkins University (2023-present)
- Recruitment Committee, Center for Language and Speech Processing, Johns Hopkins University (2023-present)